Chapter 4

Orthogonality

Homework List

Strang

- Sec 4.1: 10,11,12, 22, 28
- Sec 4.2: 1, 11, 19
- Sec 4.3: 1, 5, 28
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Tucker:

- Sec 5.3: 3, 8

Section 4.1

Orthogonality of the Four Subspaces

Orthogonality of vectors

Definition: Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal to one another if and only if their dot product is zero:

$$\vec{u} \cdot \vec{v} = 0 \leftrightarrow \vec{u}^T \vec{v} = 0$$

Fact 1: The zero vector is orthogonal to every vector, including itself: $u \cdot 0 = 0, \forall u$

Fact 2: Two non-zero vectors are orthogonal to one another if and only if they lie at right angles ($\pi/2$ rad or 90°) to each other.

Orthogonality of vector spaces

Definition:

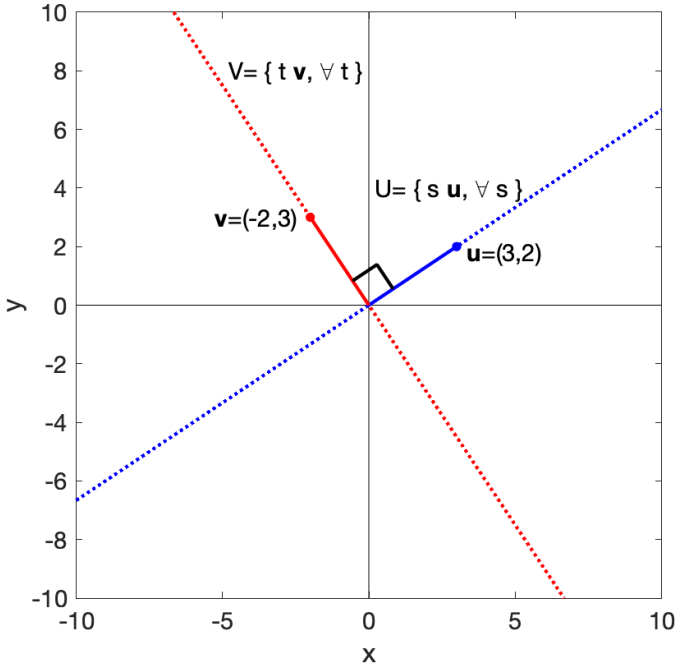
- **Two vector spaces U and V are orthogonal to one another if and only if the dot product of every vector from U with every vector from V is zero:**

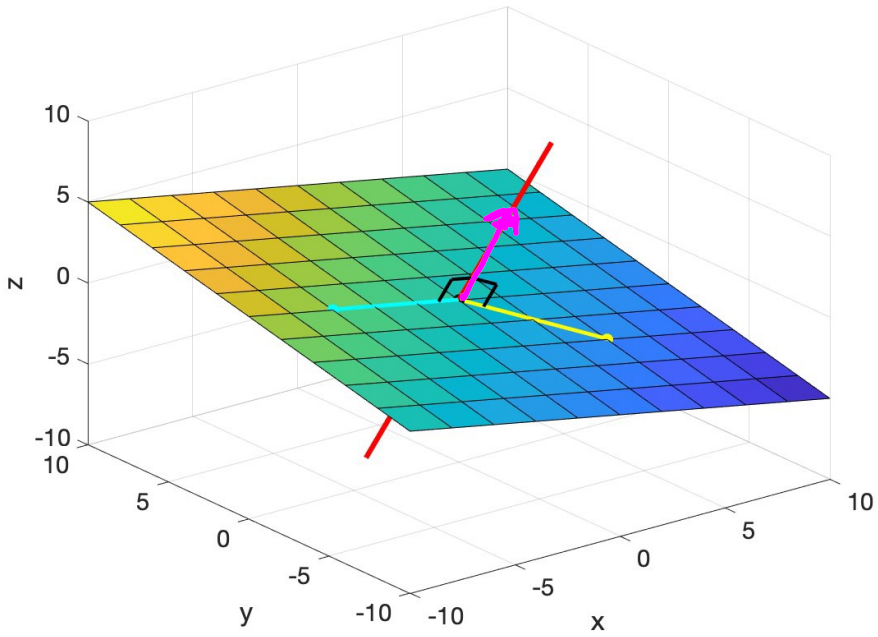
$$U \perp V \leftrightarrow \vec{u}^T \vec{v} = 0, \forall \vec{u} \in U, \forall \vec{v} \in V$$

Notation: **"$\perp$" is read as "is orthogonal to"**

Examples of orthogonal vector spaces.

- **Two lines (through the origin) that cross at a right angle.**
- **A line that intersects a plane at right angles (at the origin).**





Example

Question: Two walls V and W are perpendicular to each other. Are two subspaces (generated by two walls) orthogonal?

Answer: No! Because the meeting line is in both V and W and this line is not perpendicular to itself. Hence two planes (dimensions 2 and 2 in R3) cannot be orthogonal vector spaces.

Fact 1: The only vector that can be in two orthogonal vector spaces is 0.

Fact 2: If the vector is orthogonal to itself, then the vector is 0.

Is R(A) orthogonal to N(A)?

Claim: Every vector in N(A) must be perpendicular to each row of A.

Proof: Let x be any vector in N(A). Then

$$Ax = \begin{bmatrix} \textbf{row 1} \\ \vdots \\ \textbf{row } m \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{matrix} \longleftarrow & (\textbf{row 1}) \cdot \boldsymbol{x} \text{ is zero} \\ \\ \longleftarrow & (\textbf{row } \boldsymbol{m}) \cdot \boldsymbol{x} \text{ is zero} \end{matrix} \qquad (1)$$

The row space is orthogonal to the null space.

Alternative Proof: **Let U be the null space of a matrix, A:**

$$U = \boldsymbol{N}(A) \rightarrow A\vec{u} = \vec{0}, \forall \vec{u} \in U$$

- **Let V be the row space of A.** (Basic Proof!)
  - **V is the column space of A^T, so $A^T\vec{y} = \vec{v}$ must have a solution:**

$$V = \boldsymbol{R}(A) \rightarrow V = \boldsymbol{C}(A^T) \rightarrow \forall \vec{v} \in V, \exists \vec{y}: \vec{v} = A^T\vec{y}$$

- **Then $\vec{v}^T\vec{u}$ must be zero:** 0

$$\vec{v}^T\vec{u} = (A^T\vec{y})^T\vec{u} = (\vec{y}^T A)\vec{u} = \vec{y}^T(A\vec{u}) = \vec{y}^T\vec{0} = 0$$

$$\boldsymbol{R}(A) \perp \boldsymbol{N}(A)$$

Orthogonal complements: R(A) & N(A)

Definition:

- **Two vector spaces U and V are orthogonal complements of one another if and only if they are orthogonal and together they span the parent space:**

$$V = U^{\perp}, U = V^{\perp} \leftrightarrow U \perp V, \dim(U) \bigoplus \dim(V) = \mathbb{R}^n, U, V \subseteq \mathbb{R}^n$$

- **The row space and the null space share the same parent space:**

$$A \in \mathbb{R}^{m \times n} \rightarrow \boldsymbol{R}(A) \subseteq \mathbb{R}^n, \boldsymbol{N}(A) \subseteq \mathbb{R}^n$$

- **The row space and the null space are orthogonal:**

$$\boldsymbol{R}(A) \perp \boldsymbol{N}(A)$$

- **The dimensions of the row and null spaces sum to that of their parent space:**

$$\dim\big(\boldsymbol{R}(A)\big) + \dim\big(\boldsymbol{N}(A)\big) = \dim(\mathbb{R}^n) \leftrightarrow \operatorname{rank}(A) + \operatorname{nullity}(A) = n$$

$$\boldsymbol{R}(A) = \boldsymbol{N}(A)^{\perp}, \qquad \boldsymbol{N}(A) = \boldsymbol{R}(A)^{\perp}$$

C(A) is orthogonal complement of N(A^T).

- **The column space of A is just the row space of A^T.**
- **The column space and the left null space share the same parent space:**

$$A \in \mathbb{R}^{m\times n} \rightarrow \boldsymbol{C}(A) \subseteq \mathbb{R}^m, \boldsymbol{N}\left(A^T\right) \subseteq \mathbb{R}^m$$

- **The column space and the left null space are orthogonal:**

$$\boldsymbol{C}(A) = \boldsymbol{R}\left(A^T\right) \perp \boldsymbol{N}\left(A^T\right)$$

- **The dimensions of the column and left null spaces sum to that of their parent space:**

$$\dim\left(\boldsymbol{C}(A)\right) + \dim\left(\boldsymbol{N}\left(A^T\right)\right) = \dim(\mathbb{R}^m)$$

$$\boldsymbol{C}(A) = \boldsymbol{N}\left(A^T\right)^{\perp}, \qquad \boldsymbol{N}\left(A^T\right) = \boldsymbol{C}(A)^{\perp}$$

Fundamental Theorem of Linear Algebra

***Fundamental Theorem of Linear Algebra*, Part 1**

The column space and row space both have dimension r.

The nullspaces have dimensions $n - r$ and $m - r$.

***Fundamental Theorem of Linear Algebra*, Part 2**

$N(A)$ is the orthogonal complement of the row space $C(A^{\mathrm{T}})$ (in $\mathbf{R}^n$).

$N(A^{\mathrm{T}})$ is the orthogonal complement of the column space $C(A)$ (in $\mathbf{R}^m$).

Interesting Facts

1. The point of "complements" is that every $\boldsymbol{x}$ can be split into a *row space component* $\boldsymbol{x}_r$ and a *nullspace component* $\boldsymbol{x}_n$. When A multiplies $\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$, Figure 4.3 shows what happens to $A\boldsymbol{x} = A\boldsymbol{x}_r + A\boldsymbol{x}_n$:

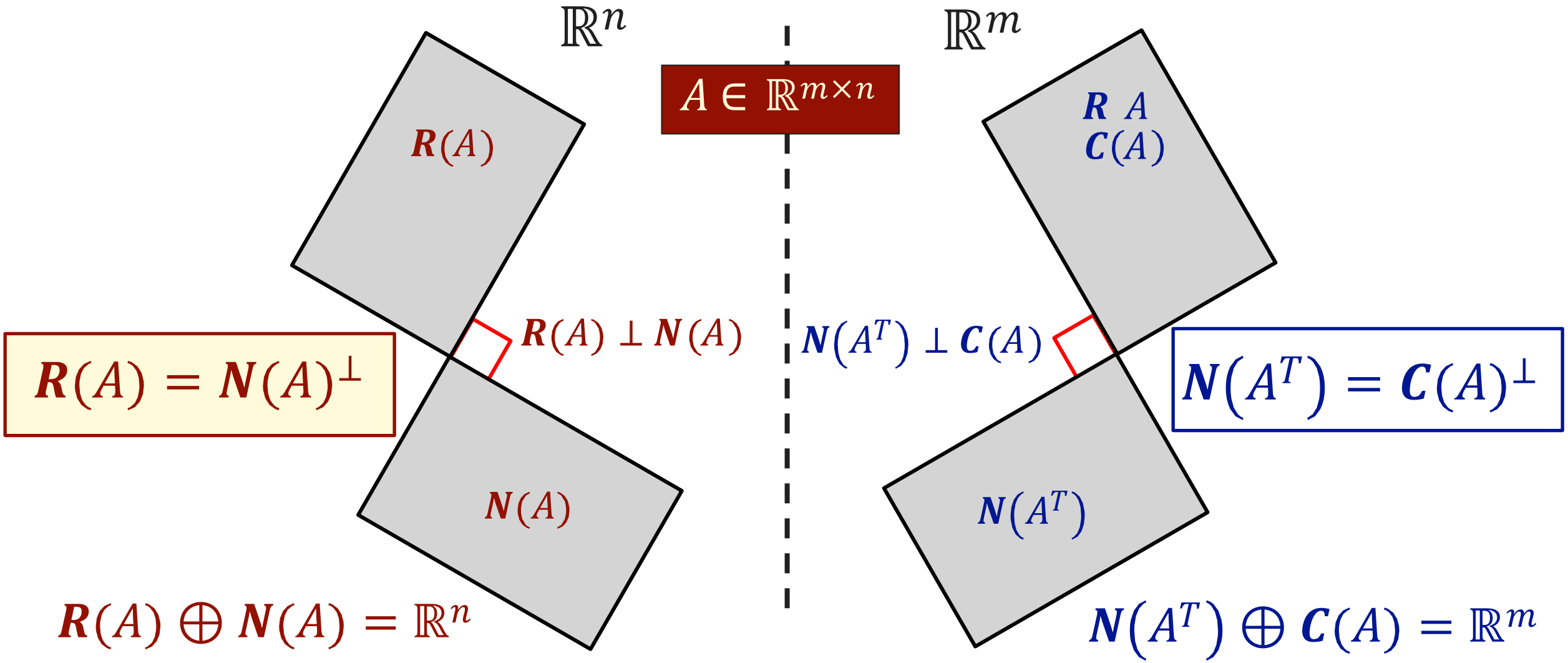
   The nullspace component goes to zero: $A\boldsymbol{x}_n = \mathbf{0}$.

   The row space component goes to the column space: $A\boldsymbol{x}_r = A\boldsymbol{x}$.

2. Every vector b in the column space comes from one and the only one vector from the row space.

More than that: *Every vector* $\mathbf{b}$ *in the column space comes from one and only one vector* $\boldsymbol{x}_r$ *in the row space*. Proof: If $A\boldsymbol{x}_r = A\boldsymbol{x}_r'$, the difference $\boldsymbol{x}_r - \boldsymbol{x}_r'$ is in the nullspace. It is also in the row space, where $\boldsymbol{x}_r$ and $\boldsymbol{x}_r'$ came from. This difference must be the zero vector, because the nullspace and row space are perpendicular. Therefore $\boldsymbol{x}_r = \boldsymbol{x}_r'$.

The four fundamental matrix subspaces.



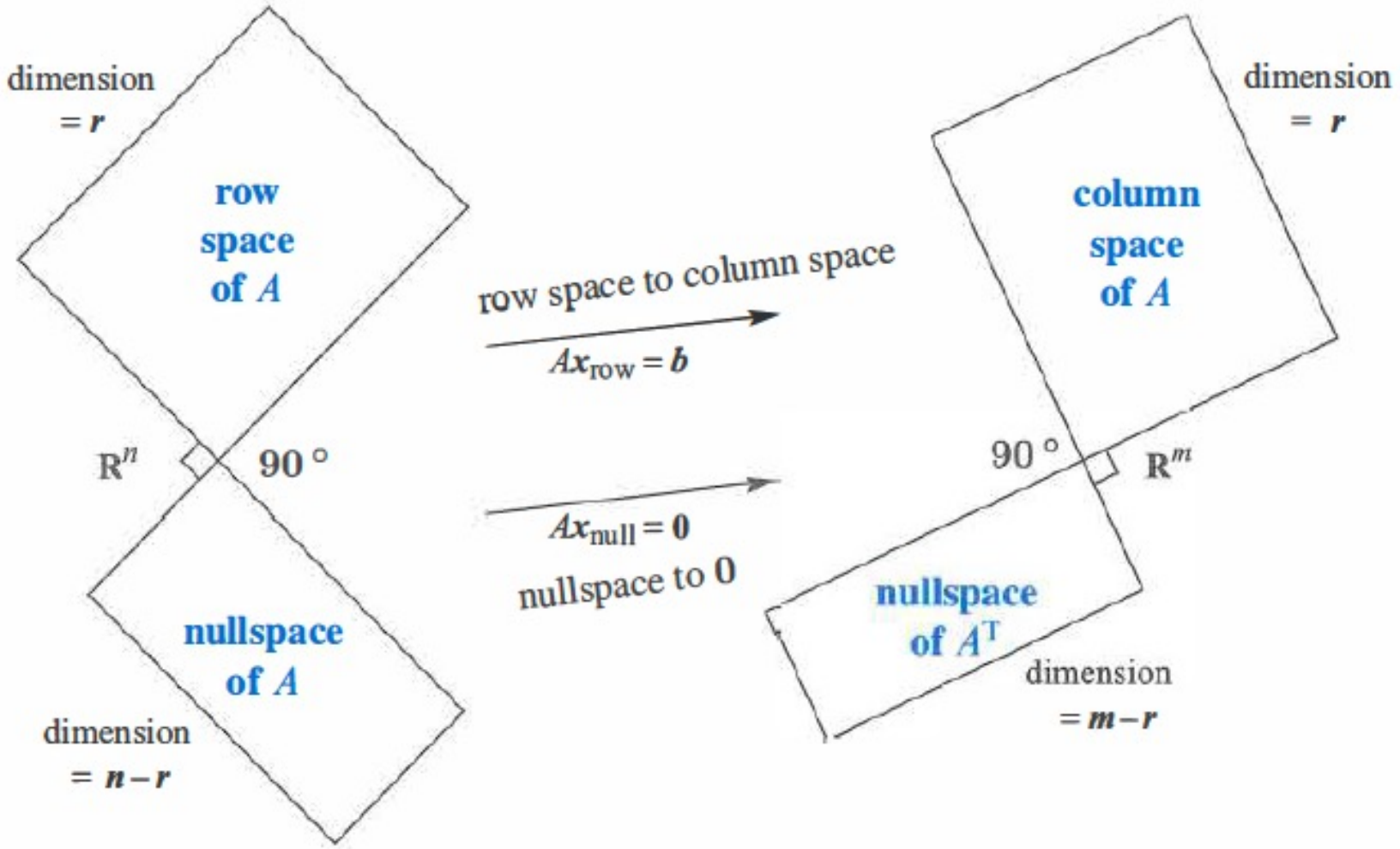


Figure 4.2: Two pairs of orthogonal subspaces. The dimensions add to n and add to m. **This is the Big Picture**—two subspaces in $\mathbf{R}^n$ and two subspaces in $\mathbf{R}^m$.

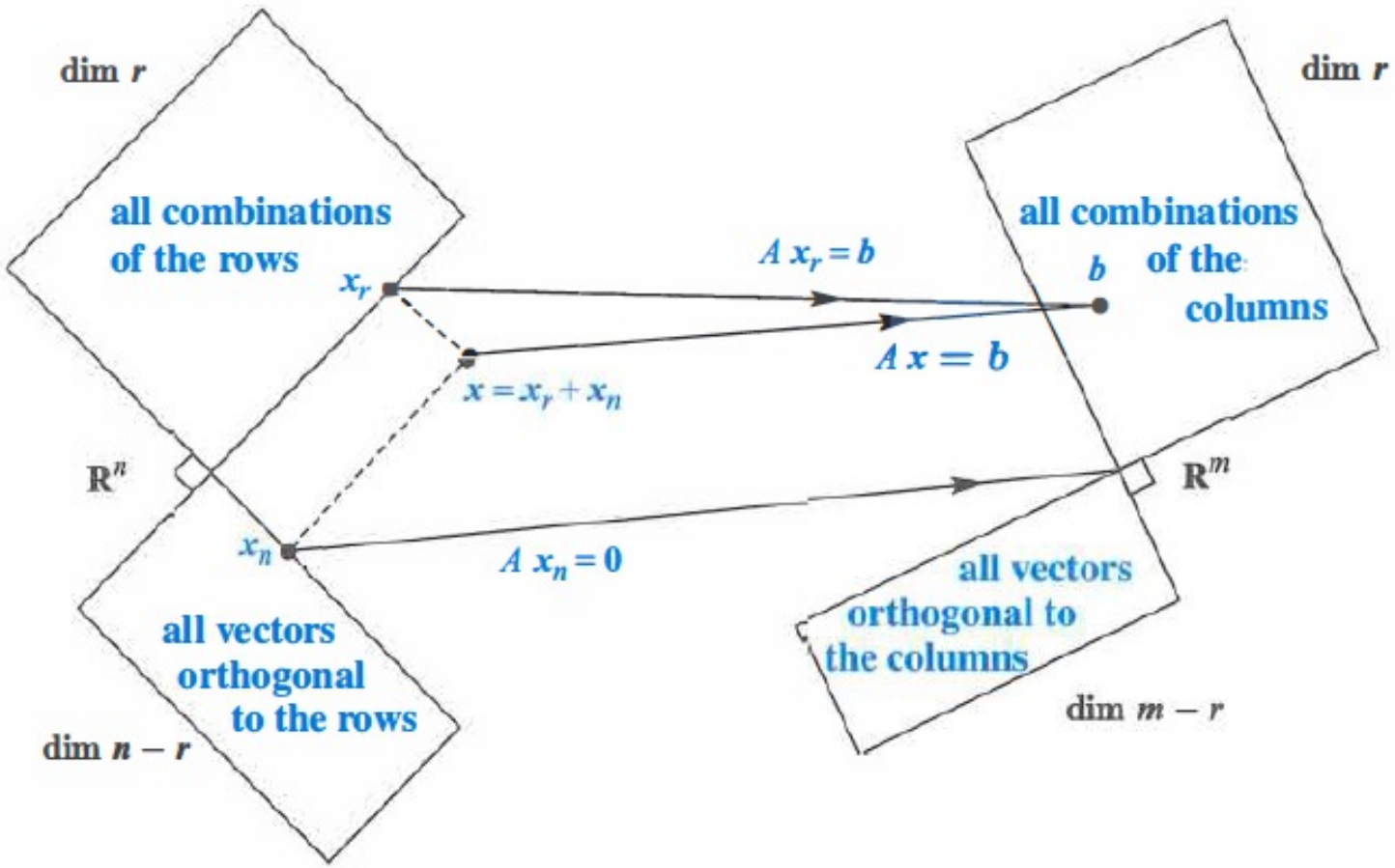


Figure 4.3: This update of Figure 4.2 shows the true action of A on $\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$. Row space vector $\boldsymbol{x}_r$ to column space, nullspace vector $\boldsymbol{x}_n$ to zero.

Basic Facts

Any n independent vectors in $\mathbf{R}^n$ must span $\mathbf{R}^n$. So they are a basis.
Any n vectors that span $\mathbf{R}^n$ must be independent. So they are a basis.

If the n columns of A are independent, they span $\mathbf{R}^n$. So $A\boldsymbol{x} = \boldsymbol{b}$ is solvable.
If the n columns span $\mathbf{R}^n$, they are independent. So $A\boldsymbol{x} = \boldsymbol{b}$ has only one solution.

Section 4.2

Projections

Projection Matrix

- What are the projections of b=(2,3,4) onto the z-axis and xy-plane?
- What are the corresponding matrices for these projections?

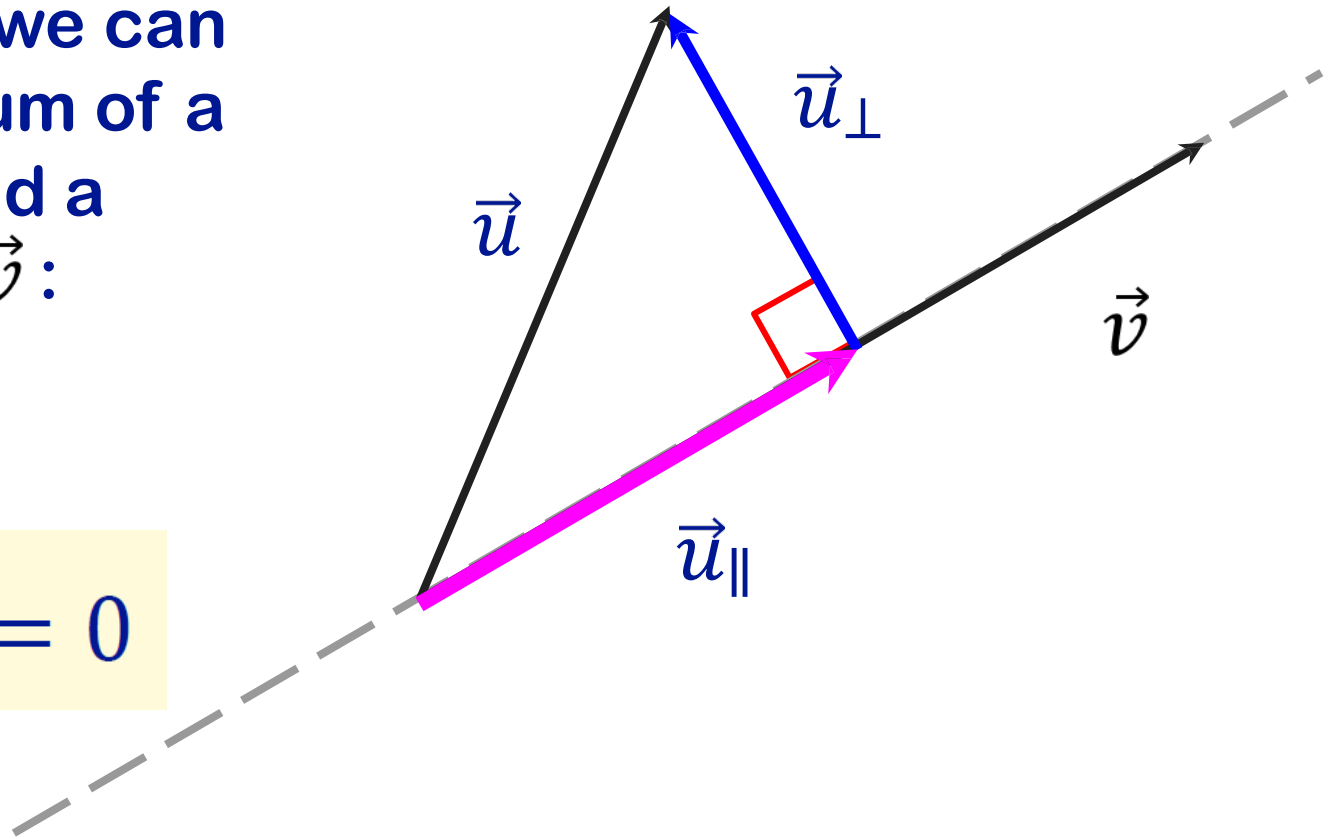
Projection matrix
Onto the z axis: $P_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix}$ Onto the xy plane: $P_2 = \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Review: projection of a vector onto a vector (line).

- **Given a two vectors, $\vec{u}$ and $\vec{v}$ we can always write $\vec{u}$ as a unique sum of a vector that is parallel to $\vec{v}$ and a vector that is orthogonal to $\vec{v}$:**

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

$$\vec{u}_{\parallel} = k\vec{v}, \qquad \vec{v}^T\vec{u}_{\perp} = 0$$



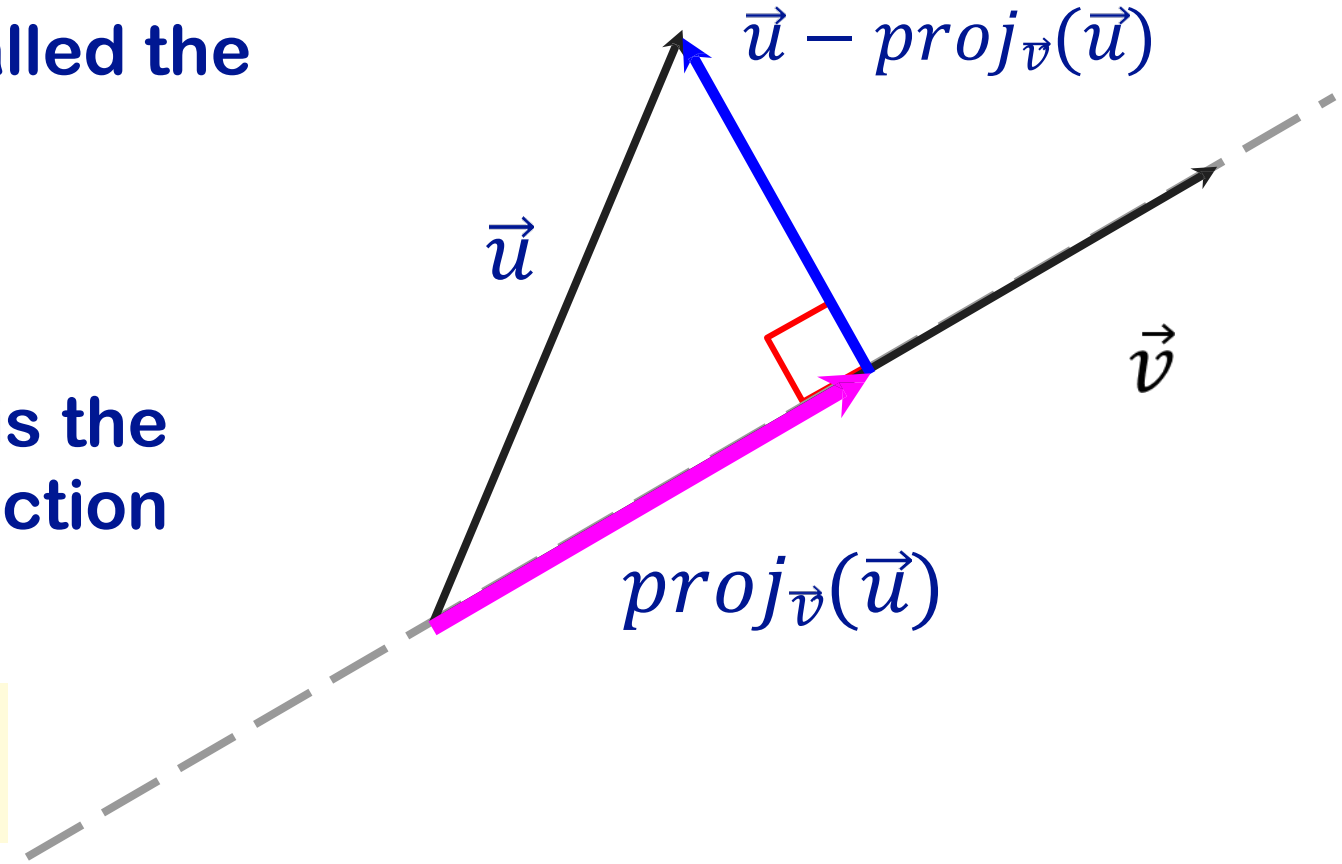
Review: projection of a vector onto a vector.

- **The parallel component is called the projection of $\vec{u}$ onto $\vec{v}$.**

$$\vec{u}_{\parallel} = proj_{\vec{v}}(\vec{u})$$

- **The orthogonal component is the difference of $\vec{u}$ and its projection onto $\vec{v}$:**

$$\vec{u}_{\perp} = \vec{u} - proj_{\vec{v}}(\vec{u})$$



Review: projection of a vector onto a vector.

- How do we find the projection?

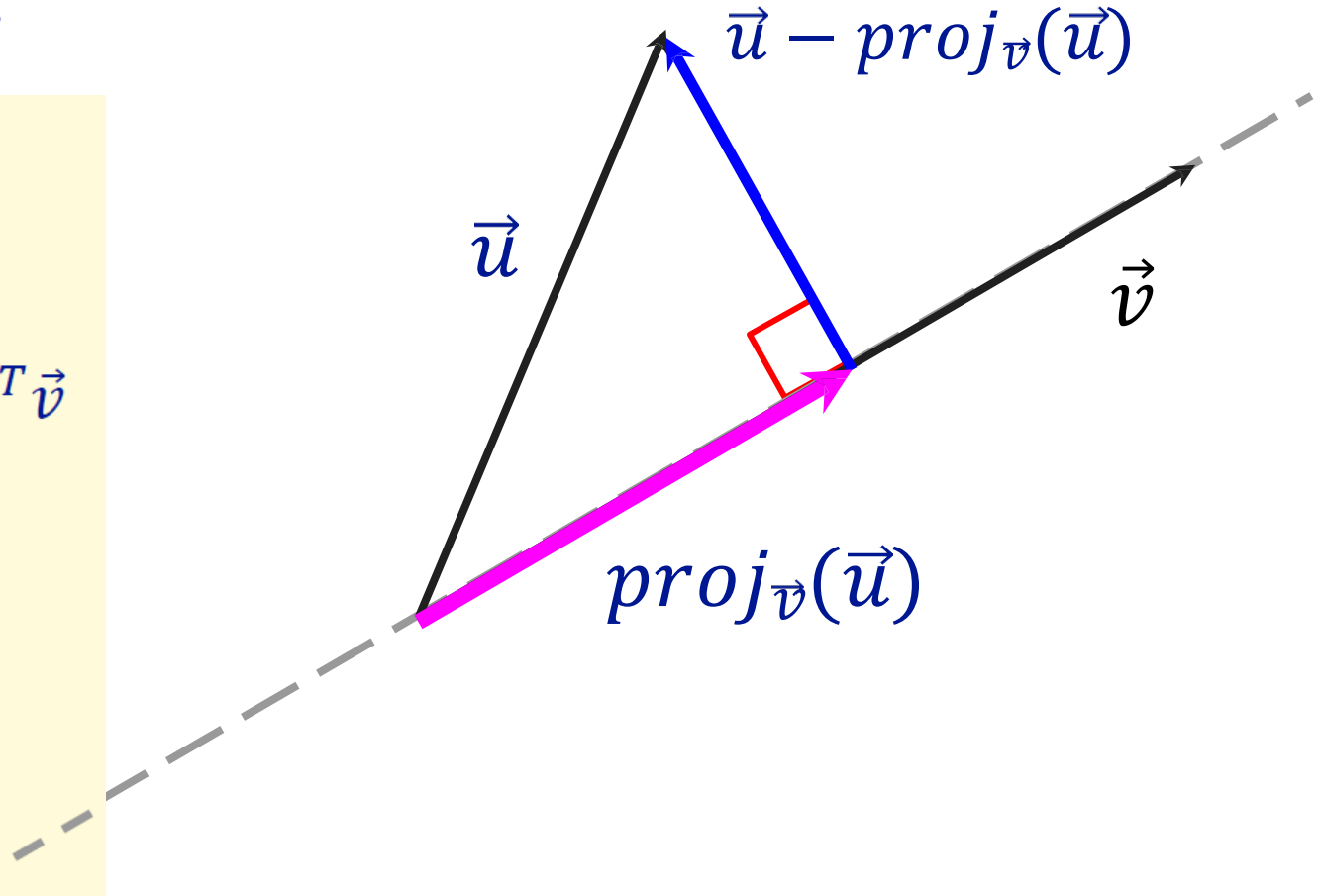
$$proj_{\vec{v}}(\vec{u}) = k\vec{v}$$

$$\vec{v}^T(\vec{u} - proj_{\vec{v}}(\vec{u})\,) = 0$$

$$\vec{v}^T(\vec{u} - k\vec{v}) = 0 \rightarrow \vec{v}^T\vec{u} = k\vec{v}^T\vec{v}$$

$$k = \frac{\vec{v}^T\vec{u}}{\vec{v}^T\vec{v}}$$

$$proj_{\vec{v}}(\vec{u}) = \left(\frac{\vec{v}^T\vec{u}}{\vec{v}^T\vec{v}}\right)\vec{v}$$



Projection matrix

The projection matrix for a vector $\vec{u}$ to a vector $\vec{v}$ is:

$$P = \frac{vv^T}{v^T v}$$

Then $P\vec{u} = \frac{vv^T u}{v^T v} = v\,\frac{v^T u}{v^T v} = proj_v^u$

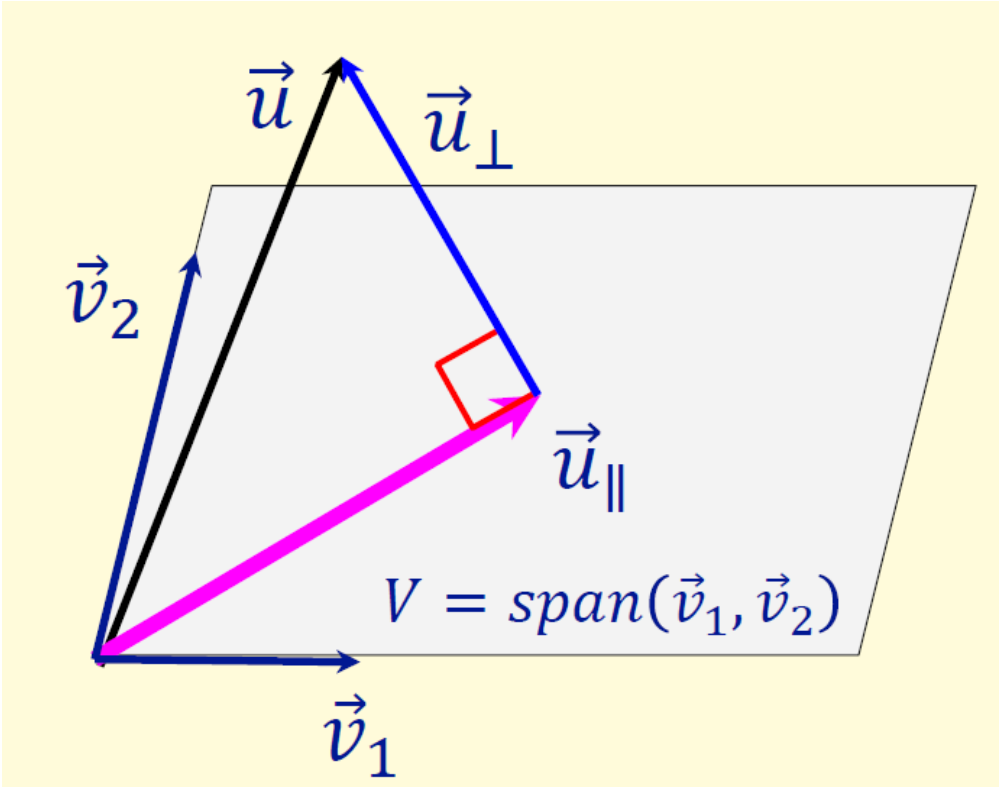
Projection of a vector onto a vector space.

- Given a vector, $\vec{u}$, and a vector space V, we can always write $\vec{u}$ as a unique sum of a vector that is in V and a vector that is orthogonal to V:

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

$$\vec{u}_{\parallel} \in V,$$

$$\vec{v}^T \vec{u}_{\perp} = 0, \forall \vec{v} \in V$$



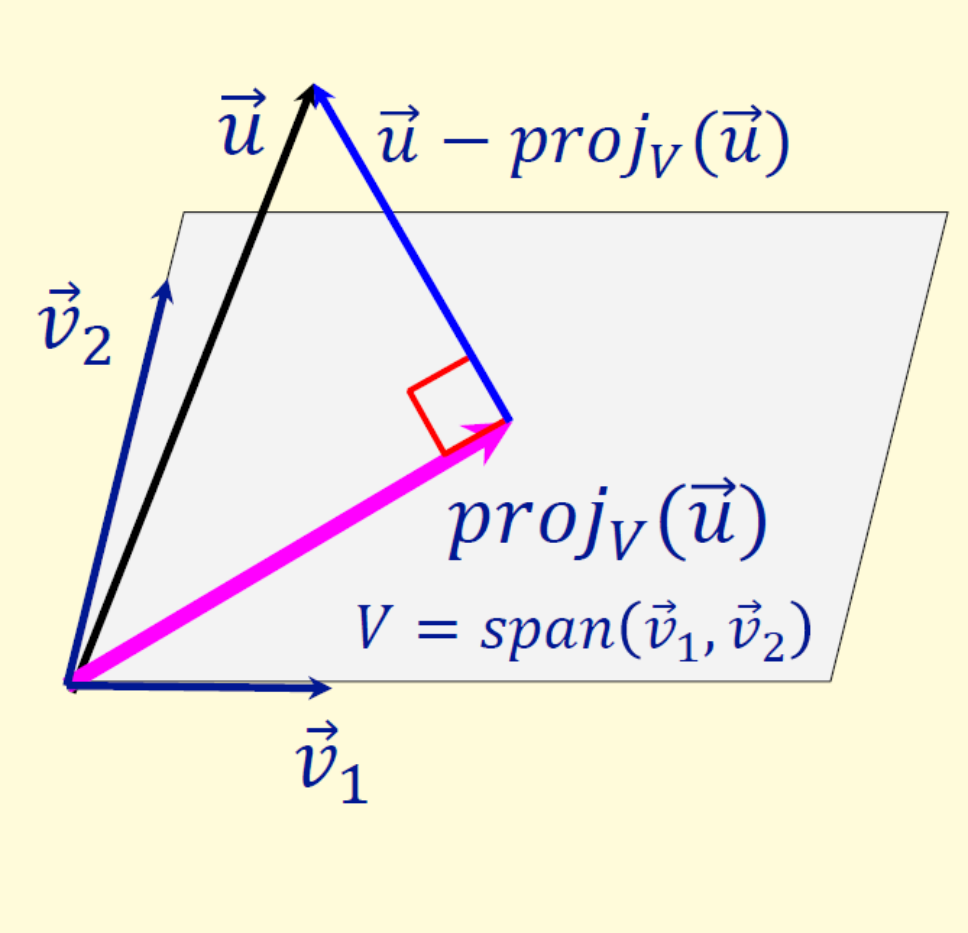
The projection of a vector onto a vector space.

- **The parallel component is called the projection of $\vec{u}$ onto V.**

$$\vec{u}_{\parallel} = proj_V(\vec{u})$$

- **The orthogonal component is the difference of $\vec{u}$ and its projection onto V:**

$$\vec{u}_{\perp} = \vec{u} - proj_V(\vec{u})$$



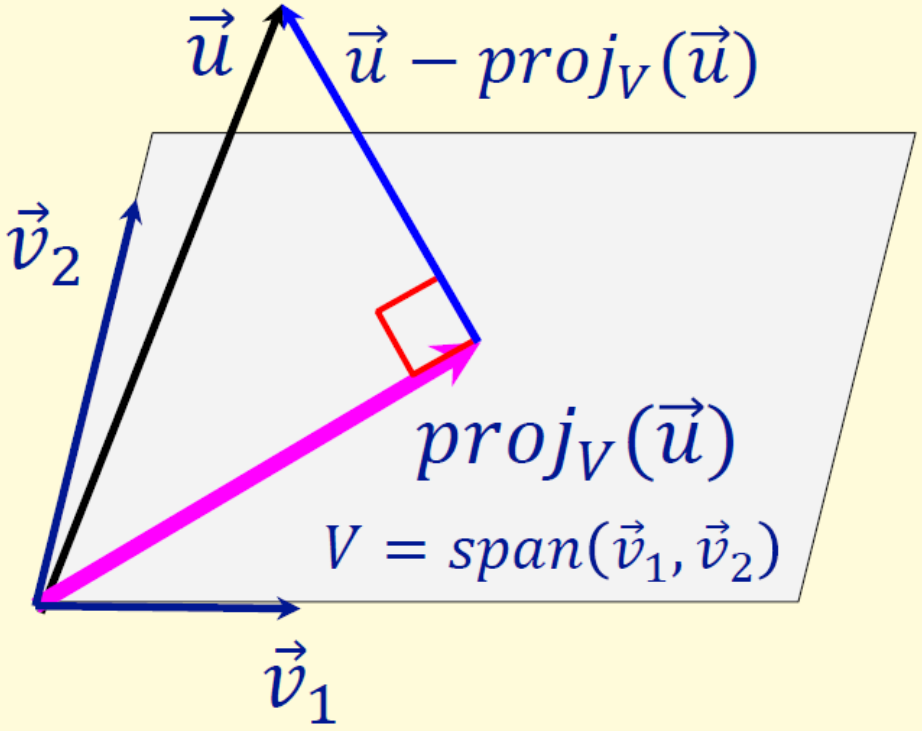
The projection of a vector onto a vector space.

- To find the projection of $\vec{u}$ onto a vector space V.
- Define a matrix A whose columns are a basis for V.
- Then:

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}$$

Note, $proj_{\vec{v}}(\vec{u})$ is just a special case:

$$proj_{\vec{v}}(\vec{u}) = \left(\frac{\vec{v}^T \vec{u}}{\vec{v}^T \vec{v}}\right) \vec{v} = \vec{v}\left(\vec{v}^T \vec{v}\right)^{-1} \vec{v}^T \vec{u}$$



The projection of a vector onto a vector space. $= C(A)$

$$A = [\vec{v}_1 \quad \vec{v}_2]$$

$$\begin{matrix} \vec{v}_1^T(\vec{u} - proj_V(\vec{u})) = 0 \\ \vec{v}_2^T(\vec{u} - proj_V(\vec{u})) = 0 \end{matrix} \rightarrow A^T(\vec{u} - proj_V(\vec{u})) = \vec{0}$$

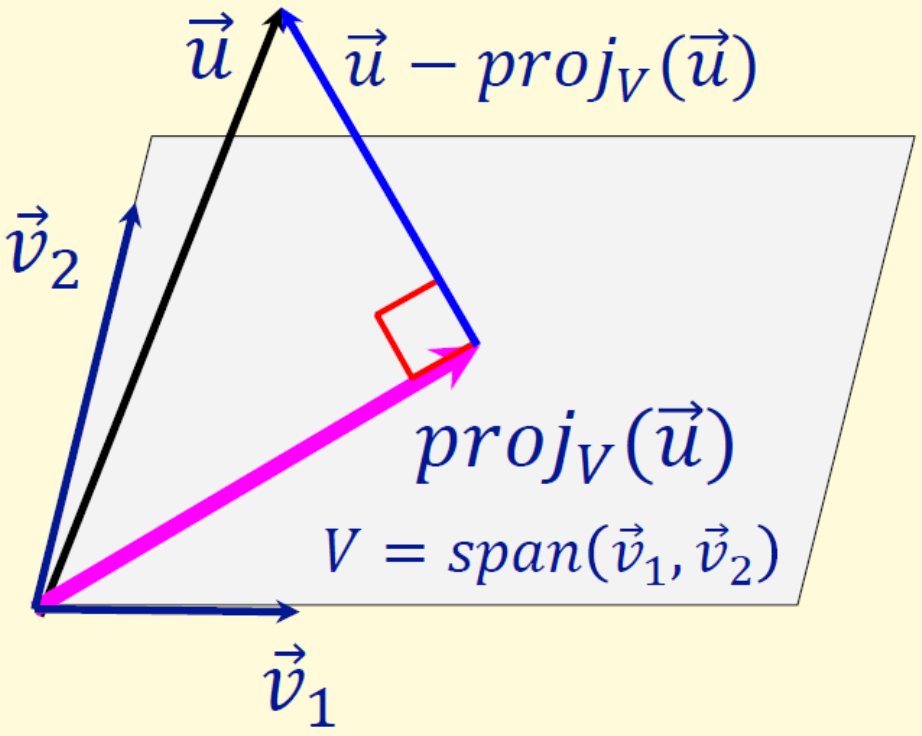
$$proj_V(\vec{u}) \in span(\vec{v}_1, \vec{v}_2) \rightarrow proj_V(\vec{u}) \in \boldsymbol{C}(A)$$

$$\exists \vec{w}: A\vec{w} = proj_V(\vec{u})$$

$$A^T(\vec{u} - proj_V(\vec{u})) = \vec{0} \rightarrow A^T\vec{u} = A^TA\vec{w}$$

$$\vec{w} = (A^TA)^{-1}A^T\vec{u}$$

$$proj_V(\vec{u}) = A(A^TA)^{-1}A^T\vec{u}$$

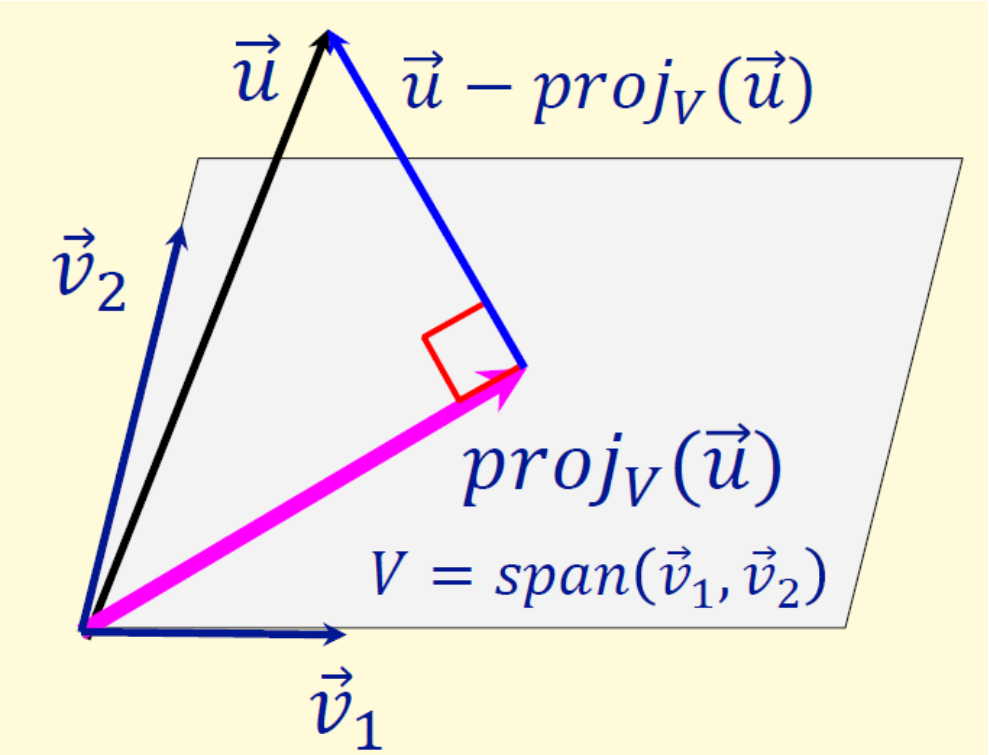


The meaning of the projection.

- The projection $proj_V(\vec{u})$ is the point in V that is closest to $\vec{u}$.

$$\min_{\vec{v}\in V}\|\vec{u}-\vec{v}\| \to proj_V(\vec{u})$$

- The orthogonal component is the deviation of $\vec{u}$ from this closest point.



Alternative way: Strang's Book

$\text{proj}_{C(A)} \vec{b}$

The combination $\boldsymbol{p} = \widehat{x}_1 \boldsymbol{a}_1 + \cdots + \widehat{x}_n \boldsymbol{a}_n = A\widehat{\boldsymbol{x}}$ that is closest to $\boldsymbol{b}$ comes from $\widehat{\boldsymbol{x}}$:

$$\textbf{Find } \widehat{\boldsymbol{x}}\,(n \times 1) \qquad A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0} \quad \text{or} \quad A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}. \tag{5}$$

This symmetric matrix $A^{\mathrm{T}}A$ is n by n. It is invertible if the $\boldsymbol{a}$'s are independent. The solution is $\widehat{\boldsymbol{x}} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$. The ***projection*** of $\boldsymbol{b}$ onto the subspace is $\boldsymbol{p}$:

$$\textbf{Find } \boldsymbol{p}\,(m \times 1) \qquad \boldsymbol{p} = A\widehat{\boldsymbol{x}} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}. \tag{6}$$

The next formula picks out the ***projection matrix*** that is multiplying $\boldsymbol{b}$ in (6):

$$\textbf{Find } \boldsymbol{P}\,(m \times m) \qquad P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}. \tag{7}$$

Strang's book

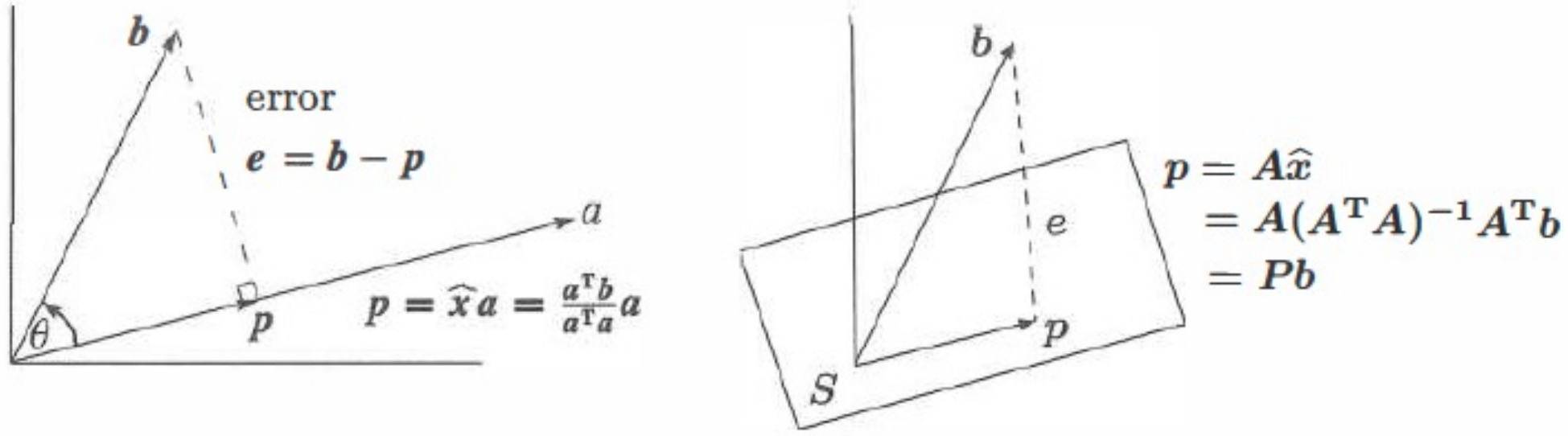


Figure 4.6: The projection p of b onto a line and onto S = column space of A.

The key step was $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \mathbf{0}$. We used geometry ($\boldsymbol{e}$ is orthogonal to each $\boldsymbol{a}$). *Linear algebra gives this "normal equation" too*, in a very quick and beautiful way:

1. Our subspace is the column space of A.

2. The error vector $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ is perpendicular to that column space.

3. Therefore $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ is in the nullspace of A^{T}! This means $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \mathbf{0}$.

$$\begin{aligned} \boldsymbol{a}_1^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) &= 0 \\ &\vdots \\ \boldsymbol{a}_n^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) &= 0 \end{aligned} \quad \text{or} \quad \begin{bmatrix} - \boldsymbol{a}_1^{\mathrm{T}} - \\ \vdots \\ - \boldsymbol{a}_n^{\mathrm{T}} - \end{bmatrix} \begin{bmatrix} \\ \boldsymbol{b} - A\widehat{\boldsymbol{x}} \\ \\ \end{bmatrix} = \begin{bmatrix} \\ \mathbf{0} \\ \\ \end{bmatrix}. \tag{4}$$

Section 4.3

Least squares approximation

Consistent and inconsistent systems

- In linear algebra, a system of linear equations is called consistent if there is at least one solution for the system. Otherwise, the linear equation system is called **inconsistent.**
- Typically, there are two kinds of inconsistent linear systems:
  - Overdetermined systems (# equations > # variables)
  - Underdetermined systems (# equations < # variables)
- What should we do if we want to solve inconsistent systems?

To repeat: We cannot always get the error $\boldsymbol{e} = \boldsymbol{b} - A\boldsymbol{x}$ down to zero. When $\boldsymbol{e}$ is zero, $\boldsymbol{x}$ is an exact solution to $A\boldsymbol{x} = \boldsymbol{b}$. *When the length of* $\boldsymbol{e}$ *is as small as possible,* $\widehat{\boldsymbol{x}}$ *is a least squares solution.* Our goal in this section is to compute $\widehat{\boldsymbol{x}}$ and use it. These are real problems and they need an answer.

Approximate solution to overdetermined system

Given an overdetermined system (# equations > # variables) $A\vec{x} = \vec{b}$ where $\vec{b}$ is not in C(A). Then the system has no solution and the echelon forms of A include zero rows.

What should we do?

Method: An approximate solution is to find the solution that yields a right-hand side as close as possible to $\vec{b}$:

$$\min_{\vec{x},\vec{p}} \left\| \vec{b} - \vec{p} \right\| \; s.t. \; A\vec{x} = \vec{p}$$

Approximate solution to overdetermined system

- **The right-hand side $\vec{p}$ for the approximate solution must be in the column space of A:**

$$\exists \vec{x}: A\vec{x} = \vec{p} \rightarrow \vec{p} \in \boldsymbol{C}(A)$$

Recall: The point closest to $\vec{b}$ in C(A) is the projection of $\vec{b}$ onto C(A):

$$\min_{\vec{p}} \left\| \vec{b} - \vec{p} \right\| \rightarrow \vec{p} = proj_{C(A)}\left(\vec{b}\right)$$

Summary: Approximate solution to overdetermined system

- **Given an inconsistent system:**

$$A\vec{x} = \vec{b}, \vec{b} \notin \boldsymbol{C}(A)$$

- **The solution that yields a right-hand side as close as possible to $\vec{b}$ is given by solving:**

$$A\vec{x} = proj_{\boldsymbol{C}(A)}\left(\vec{b}\right)$$

- **If the columns of A are linearly independent:**

$$proj_{\boldsymbol{C}(A)}\left(\vec{b}\right) = A\left(A^T A\right)^{-1} A^T \vec{b} \rightarrow A\vec{x} = A\left(A^T A\right)^{-1} A^T \vec{b}$$

$$\vec{x} = \left(A^T A\right)^{-1} A^T \vec{b}$$

Theorem: Symmetric, square & invertible

Theorem: $A^T A$ is invertible if and only if A has linearly independent columns.

Proof $A^{\mathrm{T}}A$ is a square matrix (n by n). For every matrix A, we will now show that $A^{\mathrm{T}}A$ ***has the same nullspace as*** A. When the columns of A are linearly independent, its nullspace contains only the zero vector. Then $A^{\mathrm{T}}A$, with this same nullspace, is invertible.

Let A be any matrix. If $\boldsymbol{x}$ is in its nullspace, then $A\boldsymbol{x} = \mathbf{0}$. Multiplying by A^{T} gives $A^{\mathrm{T}}A\boldsymbol{x} = \mathbf{0}$. So $\boldsymbol{x}$ is also in the nullspace of $A^{\mathrm{T}}A$.

Now start with the nullspace of $A^{\mathrm{T}}A$. **From** $A^{\mathrm{T}}A\boldsymbol{x} = \mathbf{0}$ **we must prove** $A\boldsymbol{x} = \mathbf{0}$. We can't multiply by $(A^{\mathrm{T}})^{-1}$, which generally doesn't exist. Just multiply by $\boldsymbol{x}^{\mathrm{T}}$:

$$(\boldsymbol{x}^{\mathrm{T}})A^{\mathrm{T}}A\boldsymbol{x} = 0 \quad \text{or} \quad (A\boldsymbol{x})^{\mathrm{T}}(A\boldsymbol{x}) = 0 \quad \text{or} \quad \|A\boldsymbol{x}\|^2 = 0. \tag{11}$$

We have shown: If $A^{\mathrm{T}}A\boldsymbol{x} = \mathbf{0}$ then $A\boldsymbol{x}$ has length zero. Therefore $A\boldsymbol{x} = \mathbf{0}$. Every vector $\boldsymbol{x}$ in one nullspace is in the other nullspace. If $A^{\mathrm{T}}A$ has dependent columns, so has A. If $A^{\mathrm{T}}A$ has independent columns, so has A. This is the good case: $A^{\mathrm{T}}A$ is invertible.

The pseudoinverse of a full column rank matrix

- **For any matrix, A, the columns are linearly independent if and only if A is full column rank:**

$$\text{rank}(A) = n, A \in \mathbb{R}^{m \times n}$$

- **Define a pseudoinverse:**

$$A^{+} = \left(A^{T}A\right)^{-1} A^{T}$$

Key: A might be a singular matrix!!!

- **A^{+} is a left inverse:**

$$A^{+}A = \left(A^{T}A\right)^{-1} A^{T}A = I$$

$$AA^{+} = A\left(A^{T}A\right)^{-1} A^{T} = P \neq I$$

- **For a nonsingular matrix, A^{+} reduces to the inverse:**

$$A^{+} = A^{-1}\left(A^{T}\right)^{-1} A^{T} = A^{-1}$$

The pseudoinverse of a fullcolumn rank matrix.

- **Define a pseudoinverse for full column rank matrices:**

$$\text{rank}(A) = n, A \in \mathbb{R}^{m\times n} \rightarrow A^{+} = \left(A^{T}A\right)^{-1} A^{T}$$

- **The pseudoinverse gives us the "best" answer for any right-hand side:**

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, \forall \vec{b}$$

- **The pseudoinverse gives the exact solution to $A\vec{x} = \vec{b}$ if there is one, and the solution to $A\vec{x} = \vec{p}$ with the smallest error $\left\|\vec{b} - \vec{p}\right\|$ if there is $A\vec{x} = \vec{b}$ is inconsistent.**
- **The error vector is given by:**

$$\vec{e} = \vec{b} - \vec{p} = \vec{b} - AA^{+}\vec{b} = (I - P)\vec{b}$$

Quality of the approximate solution.

- **The pseudoinverse gives the best approximate answer to an overdetermined system, where best is defined as minimizing the length of** $\vec{e} = \vec{b} - \vec{p}$

  This is the "least squares" solution, as what is minimized is the sum of the squared error for every element of b**.**

- **However, this "best" approximation is not necessarily a "good" approximation.**
  - **Error may be very large.**
  - **Approximate solution may violate real-world constraints.**
  - **The least squares approximation may not be the best for the problem.**

An approximate solutions to an underdetermined inconsistent system.

- Given an underdetermined system (# equations < # variables: either infinitely many solutions or no solutions (inconsistent system)):

$$A\vec{x} = \vec{b}, \vec{b} \notin \boldsymbol{C}(A)$$

- **The solution that yields a right-hand side as close as possible to $\vec{b}$ is given by solving:**

$$A\vec{x} = proj_{\boldsymbol{C}(A)}\left(\vec{b}\right)$$

- **What if the columns of are linearly dependent?**

$$\text{rank}(A) < n, A \in \mathbb{R}^{m\times n}$$
$$\text{rank}\left(A^T A\right) < n, \left(A^T A\right) \in \mathbb{R}^{n\times n}$$
$$proj_{\boldsymbol{C}(A)}\left(\vec{b}\right) \neq A\left(A^T A\right)^{-1} A^T \vec{b}$$

An approximate solutions to an underdetermined inconsistent system.

- Given an inconsistent system of less than full column rank:

$$A\vec{x} = \vec{b}, \vec{b} \notin \boldsymbol{C}(A), \operatorname{rank}(A) < n, A \in \mathbb{R}^{m\times n}$$

- Let $\hat{A}$, be the matrix derived from A by removing all free variable columns; the columns of $\hat{A}$ are a basis for $\boldsymbol{C}(A)$.

$$\boldsymbol{C}(A) = \boldsymbol{C}(\hat{A}) \rightarrow proj_{\boldsymbol{C}(A)}\left(\vec{b}\right) = proj_{\boldsymbol{C}(\hat{A})}\left(\vec{b}\right)$$

$$proj_{\boldsymbol{C}(A)}\left(\vec{b}\right) = \hat{A}\left(\hat{A}^T\hat{A}\right)^{-1}\hat{A}^T\vec{b}$$

- Solve the system:

$$A\vec{x} = \hat{A}\left(\hat{A}^T\hat{A}\right)^{-1}\hat{A}^T\vec{b}$$

- As A has free variables, this will be a general approximate solution!

In this section the situation is just the opposite. There are *no* solutions to $A\boldsymbol{x} = \boldsymbol{b}$. *Instead of splitting up* $\boldsymbol{x}$ *we are splitting up* $\boldsymbol{b} = \boldsymbol{p} + \boldsymbol{e}$. Figure 4.7 shows the big picture for least squares. Instead of $A\boldsymbol{x} = \boldsymbol{b}$ we solve $A\widehat{\boldsymbol{x}} = \boldsymbol{p}$. The error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is unavoidable.

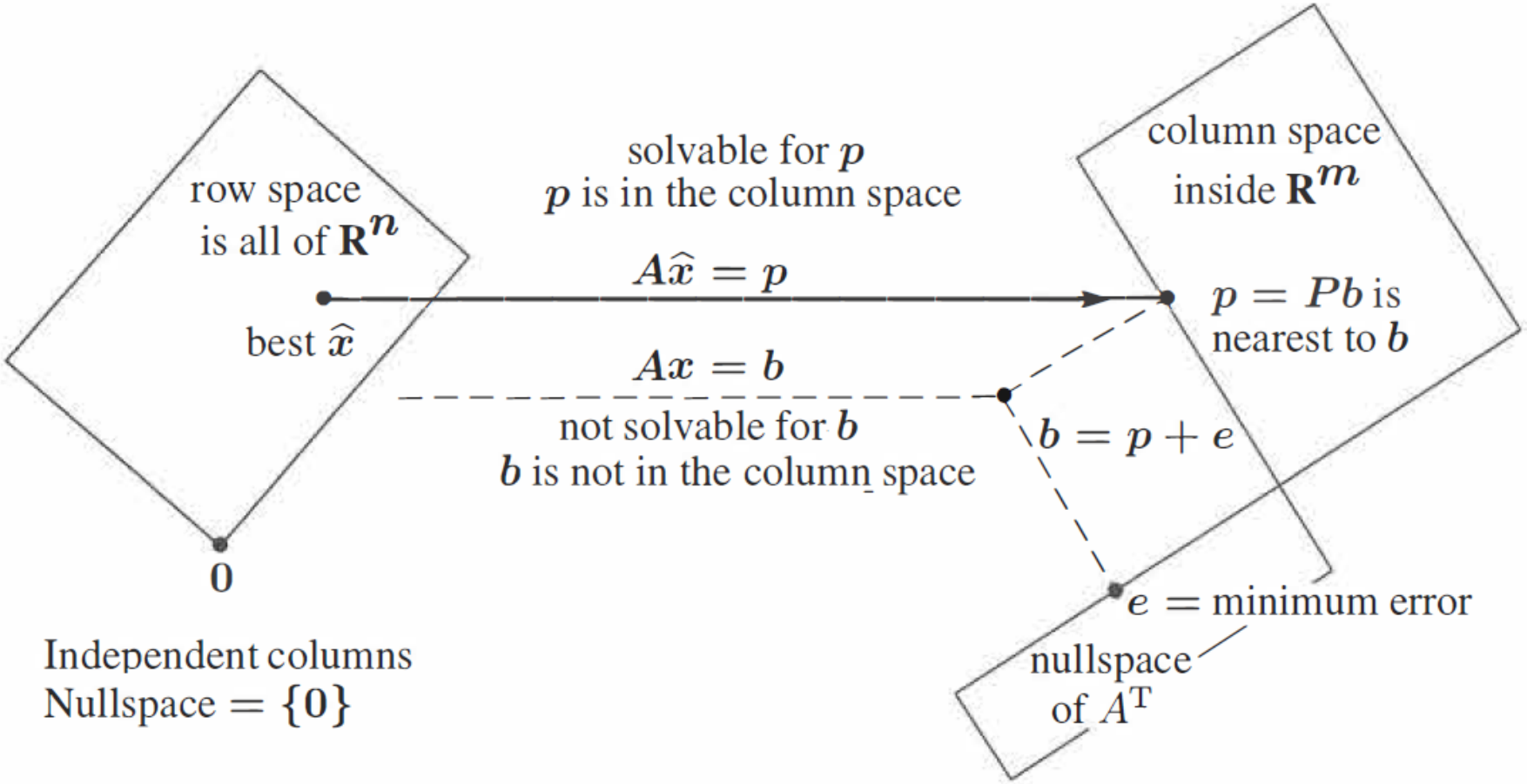


Figure 4.7: The projection $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$ is closest to $\boldsymbol{b}$, so $\widehat{\boldsymbol{x}}$ minimizes $E = \|\boldsymbol{b} - A\boldsymbol{x}\|^2$.

Example:

Example 1 A crucial application of least squares is fitting a straight line to m points. Start with three points: *Find the closest line to the points* $(0,6),(1,0)$, *and* $(2,0)$.

No straight line $b = C + Dt$ goes through those three points. We are asking for two numbers C and D that satisfy three equations: $n = 2$ and $m = 3$. Here are the three equations at $t = 0, 1, 2$ to match the given values $b = 6, 0, 0$:

$\boldsymbol{t = 0}$	The first point is on the line $b = C + Dt$ if	$C + D \cdot 0 = 6$
$\boldsymbol{t = 1}$	The second point is on the line $b = C + Dt$ if	$C + D \cdot 1 = 0$
$\boldsymbol{t = 2}$	The third point is on the line $b = C + Dt$ if	$C + D \cdot 2 = 0.$

This 3 by 2 system has *no solution*: $\boldsymbol{b} = (6, 0, 0)$ is not a combination of the columns $(1, 1, 1)$ and $(0, 1, 2)$. Read off $A, \boldsymbol{x}$, and $\boldsymbol{b}$ from those equations:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad \boldsymbol{x} = \begin{bmatrix} C \\ D \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \quad A\boldsymbol{x} = \boldsymbol{b} \text{ is } \textit{not} \text{ solvable.}$$

The same numbers were in Example 3 in the last section. We computed $\widehat{x} = (5, -3)$. **Those numbers are the best C and D, so $5 - 3t$ will be the best line for the 3 points.** We must connect projections to least squares, by explaining why $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

By calculus Most functions are minimized by calculus! The graph bottoms out and the derivative in every direction is zero. Here the error function E to be minimized is a *sum of squares* $e_1^2 + e_2^2 + e_3^2$ (the square of the error in each equation):

$$E = \|A\boldsymbol{x} - \boldsymbol{b}\|^2 = (C + D \cdot 0 - 6)^2 + (C + D \cdot 1)^2 + (C + D \cdot 2)^2. \qquad (3)$$

The unknowns are C and D. With two unknowns there are *two derivatives*—both zero at the minimum. They are "partial derivatives" because $\partial E/\partial C$ treats D as constant and $\partial E/\partial D$ treats C as constant:

$$\partial E/\partial C = 2(C + D \cdot 0 - 6) \quad + 2(C + D \cdot 1) \quad + 2(C + D \cdot 2) \quad = 0$$

$$\partial E/\partial D = 2(C + D \cdot 0 - 6)(\mathbf{0}) + 2(C + D \cdot 1)(\mathbf{1}) + 2(C + D \cdot 2)(\mathbf{2}) = 0.$$

$\partial E/\partial D$ contains the extra factors $\mathbf{0}, \mathbf{1}, \mathbf{2}$ from the chain rule. (The last derivative from $(C + 2D)^2$ was 2 times $C + 2D$ times that extra 2.) Those factors are just $\mathbf{1}, \mathbf{1}, \mathbf{1}$ in $\partial E/\partial C$.

It is no accident that those factors 1, 1, 1 and 0, 1, 2 in the derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ are the columns of A. Now cancel 2 from every term and collect all C's and all D's:

$$\begin{array}{ll} \text{The } C \text{ derivative is zero:} & 3C + 3D = 6 \\ \text{The } D \text{ derivative is zero:} & 3C + 5D = 0 \end{array} \qquad \textbf{This matrix} \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \textbf{ is } A^{\mathrm{T}}A \qquad (4)$$

Fitting Line: Polynomial Regression

- **Consider a set of data, with m data points each consisting of a pair of values:**

$$\{(x_1, y_1\ ), (x_1, y_1), , (x_m, y_m)\}$$

- **An nth-order polynomial model for the data attempts to achieve:**

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n, \forall i \in [1, m]$$

- **A matrix representation is:**

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & & x_1^n \\ 1 & x_2 & x_2^2 & & x_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_m & x_m^2 & & x_m^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- **In general $m \gg n$ (many rows, few columns): highly overdetermined.**

Fitting Line: Polynomial Regression

- **Given a set of data:**

$$\{(x_1, y_1), (x_1, y_1), , (x_m, y_m)\}$$

- **Define:**

$$A = [\vec{c}_0 \quad \vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n], \vec{c}_j = \begin{bmatrix} x_1^j \\ x_2^j \\ \vdots \\ x_m^j \end{bmatrix}, \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}$$

- **The general model is:**

$$A\vec{q} = \vec{y}$$

- **The coefficients for the minimum least-squared error model are:**

$$\vec{q} = A^+\vec{y}, A^+ = \left(A^T A\right)^{-1} A^T$$

Examples

Worked Examples using Projections and Least Squares

Ex1: Find the projection.

- **Find the projection of the vector $\vec{u}$ onto the vector space spanned by $\overrightarrow{v_1}$ and $\overrightarrow{v_2}$, where:**

$$\vec{u} = (1,2,3,4), \qquad \overrightarrow{v_1} = (1,1,1,1), \qquad \overrightarrow{v_2} = (1,2,2,1)$$

Ex 1: Find the projection.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \rightarrow A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 10 \end{bmatrix}$$

$$[A^T A \quad I] = \begin{bmatrix} 4 & 6 & 1 & 0 \\ 6 & 10 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 6 & 1 & 0 \\ 0 & 1 & -3/2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 0 & 10 & -6 \\ 0 & 1 & -3/2 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 5/2 & -3/2 \\ 0 & 1 & -3/2 & 1 \end{bmatrix} = [I \quad (A^T A)^{-1}]$$

Ex1: Find the projection.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$(A^T A)^{-1} = \begin{bmatrix} 5/2 & -3/2 \\ -3/2 & 1 \end{bmatrix}, \vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$A^T \vec{u} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \end{bmatrix}$$

$$(A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} 5/2 & -3/2 \\ -3/2 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 15 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 0 \end{bmatrix}$$

Ex 1: Find the projection.

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}, (A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} 5/2 \\ 0 \end{bmatrix}$$

$$proj_V(\vec{u}) = A\left(A^T A\right)^{-1} A^T \vec{u} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 5/2 \\ 5/2 \\ 5/2 \end{bmatrix} = \left(\frac{5}{2}\right) \vec{v}_1$$

Ex 2: Least squares solution.

- **Find the best (in a least-squares sense) approximate solution to the inconsistent system:**

$$\begin{array}{llll} x_1 & -x_2 & +x_3 & = 5 \\ x_1 & & +x_3 & = 5 \\ x_1 & +x_2 & & = 5 \\ & x_2 & +x_3 & = 5 \end{array}$$

Least squares solution.

$$\begin{array}{llll} x_1 & -x_2 & +x_3 & = 5 \\ x_1 & & +x_3 & = 5 \\ x_1 & +x_2 & & = 5 \\ & x_2 & +x_3 & = 5 \end{array} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$A\vec{x} = \vec{b} \rightarrow A\vec{x} = proj_{C(A)}\left(\vec{b}\right) = A\left(A^T A\right)^{-1} A^T \vec{b} \rightarrow \vec{x} = \left(A^T A\right)^{-1} A^T \vec{b}$$

$$\left(A^T A\right) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 3 \end{bmatrix}$$

Least squares solution.

$$\vec{x} = \left(A^T A\right)^{-1} A^T \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$[A^T A \quad I] = \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 2 & 0 & 3 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 5/3 & -2/3 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2/5 & 0 & 3/5 \end{bmatrix}$$

$$\to \begin{bmatrix} 3 & 0 & 0 & 9/5 & 0 & -6/5 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2/5 & 0 & 3/5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 3/5 & 0 & -2/5 \\ 0 & 1 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 1 & -2/5 & 0 & 3/5 \end{bmatrix} = \left[I \quad \left(A^T A\right)^{-1}\right]$$

Least squares solution.

$$\vec{x} = \left(A^T A\right)^{-1} A^T \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \\ 15 \end{bmatrix}$$

$$\vec{x} = \left(A^T A\right)^{-1} A^T \vec{b} = \begin{bmatrix} 3/5 & 0 & -2/5 \\ 0 & 1/3 & 0 \\ -2/5 & 0 & 3/5 \end{bmatrix} \begin{bmatrix} 15 \\ 5 \\ 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 5/3 \\ 3 \end{bmatrix}$$

Least squares solution.

$$A\vec{x} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5/3 \\ 3 \end{bmatrix} = \begin{bmatrix} 13/3 \\ 6 \\ 14/3 \\ 14/3 \end{bmatrix} \approx \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$\vec{e} = \begin{bmatrix} 13/3 \\ 6 \\ 14/3 \\ 14/3 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 1 \\ -1/3 \\ -1/3 \end{bmatrix}, \|\vec{e}\| = \frac{\sqrt{15}}{3} = \sqrt{\frac{5}{3}}$$

Ex 3: Find the pseudoinverse.

- **Use the pseudoinverse to solve the following system with multiple right-hand sides. Determine whether each solution is exact or approximate.**

$$A\vec{x} = \vec{b}_i,\ A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

Find the pseudoinverse.

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, A^{+} = \left(A^{T}A\right)^{-1}A^{T}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix} \rightarrow A^{T}A = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} A^{T}A & I \end{bmatrix} = \begin{bmatrix} 9 & 6 & 1 & 0 \\ 6 & 9 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 6 & 1 & 0 \\ 0 & 5 & -2/3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 6 & 1 & 0 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 9 & 0 & 9/5 & -6/5 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1/5 & -2/15 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix} = \begin{bmatrix} I & \left(A^{T}A\right)^{-1} \end{bmatrix}$$

Find the pseudoinverse.

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, A^{+} = \left(A^{T}A\right)^{-1}A^{T}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}, \left(A^{T}A\right)^{-1} = \begin{bmatrix} 1/5 & -2/15 \\ -2/15 & 1/5 \end{bmatrix}$$

$$A^{+} = \left(A^{T}A\right)^{-1}A^{T} = \begin{bmatrix} 1/5 & -2/15 \\ -2/15 & 1/5 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} -1/15 & -2/15 & 2/15 & 2/5 \\ 4/15 & 1/5 & 2/15 & -4/15 \end{bmatrix}$$

$$A^{+} = \left(\frac{1}{15}\right)\begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

$$\vec{x}_1 = A^+\vec{b}_1 = \left(\frac{1}{15}\right)\begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}\begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$A\vec{x}_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \\ 20 \\ 10 \end{bmatrix}, \vec{e}_1 = A\vec{x}_1 - \vec{b}_1 = \begin{bmatrix} 0 \\ -10 \\ +5 \\ -5 \end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix}15\\15\\15\\15\end{bmatrix}, \vec{b}_2 = \begin{bmatrix}20\\5\\30\\20\end{bmatrix}, \vec{b}_3 = \begin{bmatrix}15\\30\\30\\15\end{bmatrix}$$

$$\vec{x}_2 = A^+\vec{b}_2 = \left(\frac{1}{15}\right)\begin{bmatrix}-1 & -2 & 2 & 6\\4 & 3 & 2 & -4\end{bmatrix}\begin{bmatrix}20\\5\\30\\20\end{bmatrix} = \begin{bmatrix}10\\5\end{bmatrix}$$

$$A\vec{x}_2 = \begin{bmatrix}1 & 2\\0 & 1\\2 & 2\\2 & 0\end{bmatrix}\begin{bmatrix}10\\5\end{bmatrix} = \begin{bmatrix}20\\5\\30\\20\end{bmatrix}, \vec{e}_2 = A\vec{x}_2 - \vec{b}_2 = \begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix}15\\15\\15\\15\end{bmatrix}, \vec{b}_2 = \begin{bmatrix}20\\5\\30\\20\end{bmatrix}, \vec{b}_3 = \begin{bmatrix}15\\30\\30\\15\end{bmatrix}$$

$$\vec{x}_3 = A^+\vec{b}_3 = \left(\frac{1}{15}\right)\begin{bmatrix}-1 & -2 & 2 & 6\\4 & 3 & 2 & -4\end{bmatrix}\begin{bmatrix}15\\30\\30\\15\end{bmatrix} = \begin{bmatrix}5\\10\end{bmatrix}$$

$$A\vec{x}_3 = \begin{bmatrix}1 & 2\\0 & 1\\2 & 2\\2 & 0\end{bmatrix}\begin{bmatrix}5\\10\end{bmatrix} = \begin{bmatrix}25\\10\\30\\10\end{bmatrix}, \vec{e}_3 = A\vec{x}_3 - \vec{b}_3 = \begin{bmatrix}+10\\-20\\0\\-5\end{bmatrix}$$

Examples

Real Applications with Projections and Least Squares

Ex 1: Office workload.

A municipality has two offices that handle construction permits:

- **Each week the main office processes 600 permit applications, inspects 300 on-going projects and handles 120 complaints; the branch office processes 400 permit applications, inspects 100 on-going projects and handles 80 complaints.**
- **If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand?**
- **Is this a good solution?**
- **How would this change if the complaint load was reduced by 50%?**

Office workload.

A municipality has two offices that handle construction permits:

- **Each week, the main office processes 600 permit applications, inspects 300 on-going projects and handles 120 complaints; the branch office processes 400 permit applications, inspects 100 on-going projects and handles 80 complaints.**
- **If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.**

- **Let x_1 and x_2 be the weeks worked by the main and branch offices, respectively**
- **Write one equation for permits, one for inspections and one for complaints:**

$$\begin{aligned} 600x_1 \quad &+400x_2 \quad &= 2000 \\ 300x_1 \quad &+100x_2 \quad &= 1000 \\ 120x_1 \quad &+80x_2 \quad &= 800 \end{aligned}$$

- **Simplify each equation:**

$$\begin{aligned} 3x_1 \quad &+2x_2 \quad &= 10 \\ \rightarrow 3x_1 \quad &+1x_2 \quad &= 10 \\ 3x_1 \quad &+2x_2 \quad &= 20 \end{aligned}$$

Office workload.

- **Express in matrix form:**

$$\begin{aligned} 3x_1 \quad &+2x_2 \quad = 10 \\ 3x_1 \quad &+1x_2 \quad = 10 \\ 3x_1 \quad &+2x_2 \quad = 20 \end{aligned} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix}$$

- **Solve with the pseudoinverse**

$$\vec{x} = A^{+}\vec{b}, A^{+} = \left(A^{T}A\right)^{-1}A^{T}$$

Office workload.

$$A^T A = \begin{bmatrix} 3 & 3 & 3 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 27 & 15 \\ 15 & 9 \end{bmatrix}$$

$$[A^T A \quad I] = \begin{bmatrix} 27 & 15 & 1 & 0 \\ 15 & 9 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 27 & 15 & 1 & 0 \\ 0 & 6/9 & -5/9 & 1 \end{bmatrix} \to \begin{bmatrix} 27 & 15 & 1 & 0 \\ 0 & 1 & -5/6 & 3/2 \end{bmatrix}$$

$$\to \begin{bmatrix} 27 & 0 & 27/2 & -45/2 \\ 0 & 1 & -5/6 & 3/2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1/2 & -5/6 \\ 0 & 1 & -5/6 & 3/2 \end{bmatrix} = \begin{bmatrix} I & (A^T A)^{-1} \end{bmatrix}$$

Office workload.

$$A^{+} = \left(A^{T}A\right)^{-1}A^{T} = \left(\frac{1}{6}\right)\begin{bmatrix} 3 & -5 \\ -5 & 9 \end{bmatrix}\begin{bmatrix} 3 & 3 & 3 \\ 2 & 1 & 2 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}$$

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix}$$

$$\vec{x} = A^{+}\vec{b}$$

$$\left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}\begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} 10 \\ 30 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 5 \end{bmatrix}$$

Office workload.

A municipality has two offices that handle construction permits:

- **If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.**
- **Is this a good solution?**
- **How would this change if the complaint load was reduced by 50%?**

$$\vec{x} = A^{+}\vec{b} = \begin{bmatrix} 5/3 \\ 5 \end{bmatrix}$$

- **The main office needs to work 1.67 weeks (8-12 days) and the branch office needs to work 5 weeks (25-35 days)!**

$$A\vec{x} = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 5/3 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 10 \\ 15 \end{bmatrix} \rightarrow \begin{bmatrix} 3000 \\ 1000 \\ 600 \end{bmatrix}$$

$$\vec{e} = \begin{bmatrix} 3000 \\ 1000 \\ 600 \end{bmatrix} - \begin{bmatrix} 2000 \\ 1000 \\ 800 \end{bmatrix} = \begin{bmatrix} 1000 \\ 0 \\ -200 \end{bmatrix} \rightarrow \begin{bmatrix} +50\% \\ 0 \\ -25\% \end{bmatrix}$$

- **This isn't a great solution!**

Office workload.

A municipality has two offices that handle construction permits:

- **If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.**
- **Is this a good solution?**
- **How would this change if the complaint load was reduced by 50%?**

$$\vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix} \rightarrow \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$$\vec{x} = A^{+}\vec{b}$$

$$\left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}\begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} 10/3 \\ 0 \end{bmatrix}$$

$$A\vec{x} = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 10/3 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \vec{b}$$

- **This gives an exact solution!**
- **The main office needs to work 3.33 weeks (17-24 days) and the branch office needs to work none.**

Ex2: Regression

- **Set up, but do not solve, the following regression problems using the given data set for earnings between generations; data is average annual income for parents and their children, in thousands of dollars (and the numbers are completely made up):**

Parents	$15K	$30K	$45K	$60K	$80K	$100K	$150K	$200K	$250K
Children	$16K	$29K	$43K	$55K	$82K	$130K	$190K	$280K	$400K

- **For each case, give an expression both for the parameters of the fit, and for the total squared error; consider the parents' income to be the independent variable.**
  a) **Find the best linear fit, $y = mx + b$.**
  b) **Find the best linear fit through through the origin.**
  c) **Find the best quadratic fit, $y = ax^2 + bx + c$.**
  d) **Find the best quadratic fit through the origin.**

Regression

- **Given a set of data:**

$$\{(x_1, y_1), (x_2, y_2), , (x_m, y_m)\}$$

- **Target model:**

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n, \forall i \in [1, m]$$

$$A\vec{q} = \vec{y}, A = [\vec{c}_0 \quad \vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n], \vec{c}_j = \begin{bmatrix} x_1^j \\ x_2^j \\ \vdots \\ x_m^j \end{bmatrix}, \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}$$

$$\vec{q} = A^+\vec{y}, \; A^+ = \left(A^T A\right)^{-1} A^T$$

Best linear fit: y = mx + b

Target model: $y = a_0 + a_1 x,\ a_0 = b,\ a_1 = m$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 1 & 15 \\ 1 & 30 \\ 1 & 45 \\ 1 & 60 \\ 1 & 80 \\ 1 & 100 \\ 1 & 150 \\ 1 & 200 \\ 1 & 250 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} b \\ m \end{bmatrix} = A^+\vec{y},\ A^+ = \left(A^T A\right)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best linear fit through origin, b=0

Target model: $y = a_1 x,\ a_1 = m$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 15 \\ 30 \\ 45 \\ 60 \\ 80 \\ 100 \\ 150 \\ 200 \\ 250 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = [a_1] = [m] = A^+\vec{y},\ A^+ = \left(A^T A\right)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best quadratic fit: $y = ax^2 + bx + c$

Target model: $y = a_0 + a_1 x + a_2 x^2,\ a_0 = c,\ a_1 = b,\ a_2 = a$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 1 & 15 & 15^2 \\ 1 & 30 & 30^2 \\ 1 & 45 & 45^2 \\ 1 & 60 & 60^2 \\ 1 & 80 & 80^2 \\ 1 & 100 & 100^2 \\ 1 & 150 & 150^2 \\ 1 & 200 & 200^2 \\ 1 & 250 & 250^2 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} c \\ b \\ a \end{bmatrix} = A^+\vec{y},\ A^+ = \left(A^T A\right)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best quadratic fit through origin, c=0

Target model: $y = a_1 x + a_2 x^2$, $a_1 = b$, $a_2 = a$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 15 & 15^2 \\ 30 & 30^2 \\ 45 & 45^2 \\ 60 & 60^2 \\ 80 & 80^2 \\ 100 & 100^2 \\ 150 & 150^2 \\ 200 & 200^2 \\ 250 & 250^2 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix} = A^+\vec{y},\ A^+ = \left(A^T A\right)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$